Week 1 Interim Submission

AI-Assisted Early Risk Stratification of Adult ED Patients Using Routinely Collected Triage Data

Name: Shari Oliver
Unit: Clinical AI & Innovation Unit, Mercer General Hospital
Interim Deliverable: Three paper summaries and draft problem statement

Proposed Research Topic

My proposed research topic is **"AI-assisted early risk stratification of adult emergency department patients using routinely collected triage data"**. This topic focuses on how artificial intelligence can support triage by using information already collected at the first point of assessment, such as age, vital signs, GCS, oxygenation status and presenting complaints. The goal is not to replace the triage nurse, but to support earlier identification of patients who may be at higher risk of deterioration or urgent clinical need.

PCC Framing

Population: Adult patients presenting to the emergency department.
Concept: AI-assisted early risk stratification using routine triage data.
Context: Emergency department triage, where nurses must rapidly prioritize patients using limited early information.

Search Strategy

Databases searched: Google Scholar and PubMed

Search terms used:

("artificial intelligence" OR "machine learning" OR "AI")

AND ("emergency department" OR "emergency care")

AND ("triage" OR "risk stratification")

AND ("clinical decision support" OR "prediction model")

Google Scholar Search string:

"artificial intelligence" "emergency department" triage "risk stratification"

PubMed Search string:

("artificial intelligence"[tiab] OR "machine learning"[tiab] OR "AI"[tiab])

AND

("emergency department"[tiab] OR "emergency care"[tiab] OR "emergency medicine"[tiab])

AND

("triage"[tiab] OR "risk stratification"[tiab] OR "clinical deterioration"[tiab])

AND

("clinical decision support"[tiab] OR "prediction model"[tiab] OR "vital signs"[tiab])

Filters applied:

- Published within the last 5 years

- Peer-reviewed journal articles

- Focused on AI-assisted triage, emergency department risk prediction or clinical decision support

- Excluded vendor blogs, opinion pieces, press releases and non-healthcare AI papers

- Written in English

Paper 1 Summary

Tyler et al. (2024) - Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review

Tyler et al. looked at how artificial intelligence is being used in hospital emergency department triage. The main issue they focused on is that EDs are often overcrowded and triage staff must make quick decisions about which patients need urgent care. The authors used a scoping review method to map what recent studies have said about AI and machine learning in ED triage.

Overall, they found that AI has potential to support triage by improving prioritisation, helping with early recognition of serious cases and supporting faster decision-making. However, they also noted that many tools still need stronger clinical validation and careful implementation before they can be safely trusted in real ED workflows.

Paper 2 Summary

Da'Costa et al. (2025) - AI-Driven Triage in Emergency Departments: A Review of Benefits, Challengesand Future Directions

Da'Costa et al. reviewed the possible benefits and challenges of using AI-driven triage systems in emergency departments. The problem they addressed is that traditional triage can be affected by overcrowding, staff workload, delays and variation in human judgement. Their review discusses how AI systems can use real-time information such as vital signs, symptoms and patient history to help prioritise patients. They reported that AI could help improve patient flow, reduce waiting times and support more consistent triage decisions. At the same time, they highlighted important limitations, including poor data quality, algorithmic bias, lack of explainability, ethical concerns and the need for clinician trust before these tools can be widely adopted.

Paper 3 Summary

Araouchi and Adda (2024) - TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers

Araouchi and Adda proposed TriageIntelli, an AI-assisted triage system designed to help health centres assess and prioritise patients more effectively. The problem they focused on is that triage decisions can be difficult when staff have to quickly interpret different types of patient information, such as symptoms, vital signs and clinical indicators. Their method involved a multimodal AI system, meaning the model was designed to combine more than one type of patient data instead of relying on a single measurement. The paper suggests that using multiple data sources can make triage support more complete and potentially improve how patients are assigned to priority levels. However, the system would still need proper validation in real clinical settings, because an AI model may perform well in design or testing but still face challenges with workflow integration, data quality, staff trust and patient safety.

Early Gap Analysis

Gap 1: Limited focus on explainable nurse-facing decision support

A common gap in the literature is that many AI triage systems focus heavily on performance measures such as accuracy or prediction scores but do not always explain the decision in a way that is useful at the bedside. In triage, nurses need to know why a patient is being flagged as high risk. A risk score alone may not be enough if the system does not show the clinical reasons behind it. For a Mercer pilot, this gap could be addressed by designing a tool that gives both a risk category and a simple explanation such as low MAP, high pulse, low SpO_2, reduced GCS or another concerning clinical pattern.

Gap 2: Need for practical testing using early triage data

Another gap is that many AI triage studies use retrospective datasets or wider electronic health record information that may not be available when the patient first arrives. In real triage, staff often have to make decisions using limited early information such as vital signs, age, GCS, oxygenation status and the presenting complaint. A Mercer pilot could focus on this early stage by testing whether a simple AI-assisted triage tool can use routine triage data to identify high-risk patients without depending on complex data that nurses may not have at the first point of assessment.

Draft Problem Statement

Emergency departments need a reliable way to identify high-risk patients early but many AI triage tools are difficult to use at the bedside because they rely on complex data or do not clearly explain their decisions. At Mercer General, a 12-week pilot could test an explainable risk stratification tool that uses routinely collected triage data to assign risk levels and show the clinical reasons behind each flag. The goal is to support nurse decision-making, improve early recognition of high-risk patients and build trust in AI-assisted triage.

Preliminary Pilot Direction

The proposed pilot would develop a simple explainable triage support prototype using routinely collected triage variables such as age, SBP, DBP, MAP, pulse, temperature, respiratory rate, GCS, SpO_2, FiO_2 and presenting complaints where available. The tool would assign patients to low, moderate or high-risk categories and provide human-readable clinical flags explaining the reason for the classification. The output would be designed for nurse review rather than automatic decision-making.

References

Araouchi, Z., & Adda, M. (2024). TriageIntelli: AI-assisted multimodal triage system for health centers. *Procedia Computer Science, 251*, 430–437. https://doi.org/10.1016/j.procs.2024.11.130

Da'Costa, A., Teke, J., Origbo, J. E., Osonuga, A., Egbon, E., & Olawade, D. B. (2025). AI-driven triage in emergency departments: A review of benefits, challenges and future directions. *International Journal of Medical Informatics, 197*, 105838. https://doi.org/10.1016/j.ijmedinf.2025.105838

Tyler, S., Olis, M., Aust, N., Patel, L., Simon, L., Triantafyllidis, C., Patel, V., Lee, D. W., Ginsberg, B., Ahmad, H., & Jacobs, R. J. (2024). Use of artificial intelligence in triage in hospital emergency departments: A scoping review. *Cureus, 16*(5), e59906. https://doi.org/10.7759/cureus.59906